Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/11/2014
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Project Title:	Molecular and Cellular Mechanisms of Space Radiation-Induced Changes in Hippocampal-Dependent Behavior		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Chen, Benjamin (University Of Texas, Dallas)		
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An unavoidable consequence of many space missions is exposure to galactic cosmic radiation (GCR), which includes HZE particles like Fe, Si, and O. Estimating radiation risks to the central nervous system (CNS) is a high research priority. Here we propose extensive analysis of how a specific region of the CNS, the hippocampus, is influenced by radiation. The hippocampus is important because it contributes to spatial learning and mood regulation, which are critical for mission success. Recently the hippocampal dentate gyrus (DG) was shown to be involved in learning challenging tasks requiring fine spatial distinction ("pattern separation"). The hippocampus is also important because it is exceptionally "neuroplastic". For example, in terms of gene transcription changes, the DG is among the most responsive CNS regions to physiological and environmental stimuli. The DG is also a site of neurogenesis throughout life, and adult neurogenesis is proposed to contribute to hippocampal function in general and pattern separation in particular.

Task Description:	Understanding how space radiation influences DG structure and function will allow far better estimation of CNS radiation risks. Research already suggests that space radiation has a negative effect on hippocampal structure and function. However, it is unknown whether space radiation specifically influences DG functions like pattern separation. It is also unknown whether the diverse neural cells types within the DG are similarly influenced by space radiation or whether, for example, there is a cell-type specific change in translation or morphology. We will use recent advances in knowledge of DG function and structure and cutting-edge but proven technical approaches to answer key remaining questions about how space radiation influences the cellular, molecular, and functional integrity of the hippocampus. For example in Aim 1, using behavioral tests with which we have recently published (delayed non-match to pattern radial arm maze for fine spatial distribution; Guo et al., Nat Med 2011), we will explore our hypothesis that space radiation decreases pattern separation in a dose- and LET-dependent manner. In Aim 2, using transgenic mice we have developed that allow inducible YFP labeling of neural stem cells and their progeny (nestin-CreERT2/R26R mice; Lagace et al., J Neurosci 2007), we will explore our hypothesis that space radiation negatively impacts early postnatally-generated DG neurons. In Aim 3, using transgenic mice developed by our colleagues that allow profiling of translating RNA in genetically-defined DG neural cells, we will explore our hypothesis that space radiation causes discrete changes in DG neural cell types in a dose- and LET-dependent manner. Unlike currently used approaches to assess total RNA profiles from cellularly heterogeneous samples (e.g. laser capture dissection and subsequent microarray analysis), this approach allows purification of specific cell types via ribosome tag and a quantitative comparison of translating RNA profiles in genetically-distinct DG cells. Importantly, all mice will be 6-7 month old at time of radiation exposure in order to be approximately the age equivalent to a 30 year old human, and thus to better assess the impact of ground-based space radiation on the fully mature brain. Tissue from Aims 1-3 will be examined using immunohistochemistry, epifluorescent and confocal microscopy, stereologic cell counting, and morphological analysis, and tissue from Aim 3 will be explored using microarrays and RNAseq on the translating RNA pools and qRT-PCR. Taken together, these aims will allow us to test our overall hypothesis that radiation will have distinct, long-lasting effects on the behavioral, cellular, and molecular integrity of the hippocampus. Thus, these results will expedite the quest to estimate CNS radiation risks due to GCR exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is critical to identify factors that regulate normal brain function and structure. Our proposed research will elucidate how ground-based space radiation influences hippocampal structure and function. The knowledge derived from the proposed experiments will provide insight into the mechanisms contributing to brain and hippocampal function and their disruption by space radiation, and thus has the potential to shed light on how we can estimate CNS risk from space radiation. This work is important given the long-duration space missions on the former Mir Space Station, the present International Space Station, and the potential exploratory missions to the Moon and Mars. Mice will be used as the primary source of nervous tissue for experimentation since most of the relevant literature concerns studies of tissues of rodent origin, and since the mouse is an excellent model for human exposure to space radiation given their remarkably similar brain structure and function and genomic similarity. At this time, using computer modeling to understand the effects of space radiation on behavioral, cellular, and molecular indices of the hippocampus is just not possible given the complexity of the CNS. Cell culture systems have proven extremely useful to ask mechanistic questions about the effects of space radiation on the hippocampus, but they are unable to address the effects of space radiation on the whole organism, particularly behavior. The combination of approaches used here - analysis of behavior, cellular morphology, and translational profile - will have significant benefits for the fields of neuroscience in general and the fields of learning and memory, stem cell biology, and even neural development in particular, but are all inspired by the need to gain greater insight into whether the CNS alterations experienced by astronauts may compromise mission success. The particular focus of this application - hippocampal dysfunction - is evident in a range of psychiatric and neurologic disorders, including Alzheimer’s Disease, depression, and addiction. Astronauts may be particularly at risk for delayed hippocampal pathological signs, as it is estimated that even just 1 year in a low-Earth orbit mission may result in 91% of hippocampal cells and 25% of hippocampal cell-nuclei hit by high LET particles. These studies will help us predict neurodegenerative or cognitive changes in astronauts during the mission (which would compromise mission success) or even after mission completion. We also focus on the hippocampus since it is exceptionally "neuroplastic" – being among the most responsive brain regions in terms of gene transcription after physiological manipulation and stimuli and being a site of the birth of new neurons throughout life. This exceptional neuroplasticity suggests that the hippocampus may be among the first brain regions to reveal a response to radiation. To best study the influence of ground-based space radiation on hippocampal structure and function in a way that is most translational from rodents to humans, we will be specifically using older mice (e.g. ~6-7months old at time of irradiation, which is equivalent to 30 human years). Aside from the general benefits of this broader work for neuroscience and humans, there are several innovative aspects to this work. For example, using mice that we have generated that allow inducible labeling and tracking of embryonic vs. adult-generated neurons, we will be able to for the first time tease apart effects on these two populations. In addition, using novel mice that allow profiling of translating RNA in genetically-defined DG neural cells (e.g. adult-generated vs. embryonic-generated DG neurons), we will make a quantitative comparison of translating RNA profiles in genetically-distinct DG neural cells in a time-, dose-, and LET-dependent manner. Utilization of these novel technologies have the implication to expand our broader understanding of mechanisms underlying neuroplasticity and even neurodegenerative processes.

Task Progress:

Summary of progress through Year 3 (11/2013 – 11/2014) of 4 years of funding.
As highlighted in the numbered findings below each of our 3 specific aims, we have met or exceeded all goals for this 3rd year.

Aim 1: Determine the impact of low- and high-LET radiation on hippocampal function.

Building off of the second year's work we have used our behavioral expertise to show that context discrimination fear conditioning (CDFC) is differentially influenced in mice that are young adult vs. mature at the time of irradiation. Specifically, mice that are mature at the time of irradiation can learn more consistently and sooner to discriminate two contexts, while mice that are younger at the time of irradiation generally show decreased context separation. We find this is true for both 28Si and 56Fe, with some dose- and delivery-dependency.

We also find that contextual fear conditioning (CFC) is differentially influenced by the age of the mice at the time of irradiation. For example, in generally mice that are young adult at the time of irradiation show diminished CFC, while mice that are mature at the time of irradiation show no difference. We find this is true for both 28Si and 56Fe, although again there is some dose- and delivery-dependency.

We find no gross difference in other behavioral and physiological measures, including weight, locomotor activity, exploration of a novel environment or dark vs. light environments, and pain threshold.

We have found that hippocampal neurogenesis is decreased in most irradiated mice, regardless of the age at the time of irradiation. This disproves our initial hypothesis that space radiation-induced improvement in CDFC was reflective of increased hippocampal neurogenesis (or less of a decrease in hippocampal neurogenesis relative to controls) in mice that were mature at the time of irradiation.

Aim 2: Determine how radiation influences the dendritic morphology of embryonic- vs. adult- generated granule cell neurons.

We have collected multiple cohorts of Nestin-CreERT2/R26R-YFP mice that received Tamoxifen at postnatal day 3 and were irradiated in adulthood. These tissues are being currently examined for the effect of both 28Si and 56Fe on dendritic morphology in cells that were generated in the early postnatal period.

Aim 3: Define the impact of radiation on the translational profile of genetically defined DG neural cells.

We have collected multiple cohorts of BAC-TRAP mice that were irradiated in adulthood. These tissues are being currently examined for the effect of both 28Si and 56Fe on RNAseq and microarray expression.

Publications

We made excellent progress on publications, as we already have 11 articles published, and at least 7 more in preparation.

Publications linked to these NASA grants:

1. Decarolis NA, Mechanic M, Petrik D, Carlton AF, Malhotra S, Ables JL, Bachoo R, Lagace DC, Eisch AJ. In vivo contribution of nestin- and GLAST-lineage cells to adult hippocampal neurogenesis. Hippocampus. 23: 708–719. doi: 10.1002/hipo.22130 (2013). PMID 23554226, PMC 3718551.

2. Eisch AJ, Petrik D. Neurogenesis and depression: a road to remission? Science. (2012) Oct 5;338(6103):72-5. doi: 10.1126/science.1222941. PMID 23042885, PMC 3756889.

3. Petrik D, Lagace DC, Eisch AJ. The neurogenesis hypothesis of affective and anxiety disorders: are we mistaking the scaffolding for the building? Neuropharmacology. Jan;62(1):21-34. Epub 2011 Sep 19 (2012) PMID 21945290, PMC 3698048.

4. Petrik D, Yun S*, Latchney SE*, Kamrudin S, LeBlanc JA, Bibb JA, Eisch AJ. Early postnatal in vivo gliogenesis from nestin-lineage progenitors requires Cdk5. PLoS One, Aug 26;8(8):e72819. doi: 10.1371/journal.pone.0072819. (2013) PMID 23991155, PMC 3753242. *authors contributed equally to this work.

5. Ninkovic J, Steiner-Mezzadri A, Jawerka M, Akinci U, Masserdotti G, Petricca S, Fischer J, von Holst A, Beckers J, Lie CD, Petrik D, Miller E, Tang J, Wu J, Lefebvre V, Demmers J, Eisch AJ, Metzger D, Crabtree G, Irmler M, Poot R, Götz M. The BAF complex interacts with Pax6 in adult neural progenitors to establish a neurogenic cross-regulatory transcriptional network. Cell Stem Cell. (Epub 2013 Aug 8) Oct 3;13(4):403-18. doi: 10.1016/j.stem.2013.07.002 (2013). PMID 23933087, PMC 4098720.

6. Rivera PD*, Shih H-Y*, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Lucero MJ, DeCarolis NA, Chen BPC, Eisch AJ. Acute and Fractionated Exposure to High-LET 56Fe HZE Particle Radiation Both Result in Similar Long-Term Deficits in Adult Hippocampal Neurogenesis. Radiation Research. (Epub Dec 9 2013) Dec;180(6):658-67. doi: 10.1667/RR13480.1. (2013) PMID 24320054, PMC 3930069.*authors contributed equally to this work.

7. Latchney SE, Masiulis I, Zaccaria KJ, Lagace DC, Powell CM, McCasland JS, Eisch AJ. Developmental and adult GAP-43 deficiency in mice dynamically alters hippocampal neurogenesis and mossy fiber volume. Developmental Neuroscience. doi: 10.1159/000357840. (2014). PMID 24576816, PMC 4035301.

8. Latchney SE, Rivera PD, Mao XW, Ferguson VL, Bateman TA, Stodieck LS, Nelson GA, Eisch AJ. The effect of spaceflight on mouse olfactory bulb volume, neurogenesis, and cell death indicates the protective effect of novel environment. Journal of Applied Physiology. 2014 Apr 17. [Epub ahead of print] PMID 24744382, PMC Journal - In Process (Publisher Provided).

9. Walker AK, Rivera PD, Wang Q, Chuang JC, Tran S, Osborne-Lawrence S, Estill SJ, Starwalt R, Huntington P, Morlock L, Naidoo J, Williams NS, Ready JM, Eisch AJ, Pieper AA, Zigman JM. The P7C3 class of neuroprotective compounds exerts antidepressant efficacy in mice by increasing hippocampal neurogenesis. Molecular Psychiatry. 2014 Apr 22. doi: 10.1038/mp.2014.34. [Epub ahead of print]. PMID 24751964, PMC in process (NIHMSID 595465).

10. DeCarolis NA*, Rivera PD*, Ahn HF, Shih H-Y, Amaral WZ, LeBlanc JA, Malhotra S, Mukherjee S, Chen BPC, Eisch AJ. 56Fe Particle Exposure Results in Long-Lasting Genomic Instability and Transiently Suppresses Adult Hippocampal Neurogenesis In Vivo. Life Sciences in Space Research. doi: 10.1016/j.lssr.2014.06.004. PMC in process

(NIHMSID 611428).*authors contributed equally and are listed alphabetically

11. Latchney SE and Eisch AJ. Therapeutic application of neural stem cells and adult neurogenesis for neurodegenerative disorders: regeneration and beyond. European Journal of Neurodegenerative Diseases. In press.

Articles in revision or preparation:

1. Palchik G*, DeCarolis NA*, Ahn HF, Lagace DC, Ables JL, Redfield RL, Cole MG, Chen BPC, Eisch AJ. Running after irradiation ameliorates the proliferation deficit but not the nestin-expressing Type-1 cell deficit in the adult hippocampal subgranular zone. In revision for resubmission to Hippocampus.*authors contributed equally to this work.

2. Lagace DC, Petrik D, Donovan MH, Noonan MA, Reister R, Farnbauch LF, Riethmacher D, Gershenfeld H, Eisch AJ. Depressive phenotype resulting from mild ablation of adult neuorgenesis by nestin-CreERT2-mediate expression of diptheria toxin fragment A in mice. In preparation for submission to Neuroscience Letters.

3. Rivera PD*, Shih H-Y*, LeBlanc JA, Cole MG, Amaral WZ, Malhotra S, DeCarolis NA, Chen BPC, Eisch AJ. High LET 28Si particle radiation leads to long-term deficits in hippocampal neurogenesis. In progress for submission to Radiation Research.

4. Lucero MJ, Ito N, Redfield RL, Ho M, Mukherjee S, Shih H-Y, Chen BPC, Birnbaum SB, Eisch AJ. HZE particle radiation improves pattern separation in older, but not younger, mice. In progress.

5. Yun S, Lucero MJ, Ito N, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. Molecular basis for space radiation-induced deficits hippocampal function in aged animals. In preparation.

6. Cole MG, Shih H-Y, LeBlanc JA, Rivera PD, Chen BPC, Eisch AJ. Short and long term effects of 16O radiation on adult hippocampal neurogenesis and indices of genomic instability. In progress.

7. Cole MG, Shih H-Y, LeBlanc JA, Chen BPC, Eisch AJ. A model of acute space radiation exposure (12C) and adult hippocampal neurogenesis. In progress

We have also presented six oral/poster presentations at conferences this past year, including:

1. Eisch AJ “CNS and Radiation Effects.” Invited Topical Review presented at The Annual Meeting of the Radiation Research Society, New Orleans, LA, September 2013.

2. Lucero M. J. Redfield R. L. Ito N. Mukherjee S. Shih H. S. Rivera P. D. Birnbaum S. G. Chen B. P. C. Eisch A. J. “Pattern Separation is Enhanced in HZE-Irradiated Adult Mice”. NASA 2013 Human Research Program Investigators’ Workshop, Galveston TX, February 2014.

3. Eisch A. J. Oral presentation: “Pattern Separation is Enhanced in HZE-Irradiated Adult Mice”. NASA 2013 Human Research Program Investigators’ Workshop, Galveston TX, February 2014.

4. Eisch AJ “CNS and Radiation Effects.” Invited Topical Review presented at The Annual Meeting of the Radiation Research Society, New Orleans, LA, September 2013.

5. Lucero MJ, Redfield RL, Ito N, Mukherjee S, Shih HS, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. “HZE particle radiation enhances pattern separation in adult mice” The Annual Meeting of the Radiation Research Society, Las Vegas, NV, September 2014.

6. Lucero MJ, Redfield RL, Ito N, Mukherjee S, Shih HS, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. “Space radiation improves pattern separation in older mice without influencing gross hippocampal function” The Annual Meeting of the Society for Neuroscience, Washington, D.C. November 2014.

Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Ninkovic J, Steiner-Mezzadri A, Jawerka M, Akinci U, Masserdotti G, Petricca S, Fischer J, von Holst A, Beckers J, Lie CD, Petrik D, Miller E, Tang J, Wu J, Lefebvre V, Demmers J, Eisch AJ, Metzger D, Crabtree G, Irmler M, Poot R, Götz M. "The BAF complex interacts with Pax6 in adult neural progenitors to establish a neurogenic cross-regulatory transcriptional network." Cell Stem Cell. 2013 Oct 3;13(4):403-18.http://dx.doi.org/10.1016/j.stem.2013.07.002 ; PMID: 23933087, PubMed Central PMCID: PMC4098720 , Oct-2013
Articles in Peer-reviewed Journals	Rivera PD*, Shih H-Y*, LeBlanc JA, Cole MG, Amaral WZ, Mukherjee S, Zhang S, Lucero MJ, DeCarolis NA, Chen BPC, Eisch AJ. (*authors contributed equally to this work.) "Acute and fractionated exposure to high-LET (56)Fe HZE-particle radiation both result in similar long-term deficits in adult hippocampal neurogenesis." Radiation Research. 2013 Dec;180(6):658-67. http://dx.doi.org/10.1667/RR13480.1 ; PubMed PMID: 24320054; PubMed Central PMCID: PMC3930069 (ED. NOTE: previously report as "in press" in Sept 2013 Task Book report) , Dec-2013
Articles in Peer-reviewed Journals	Latchney SE, Masiulis I, Zaccaria KJ, Lagace DC, Powell CM, McCasland JS, Eisch AJ. "Developmental and adult GAP-43 deficiency in mice dynamically alters hippocampal neurogenesis and mossy fiber volume." Developmental Neuroscience. 2014 Mar;36(1):44-63. http://dx.doi.org/10.1159/000357840 ; PubMed PMID: 24576816 , Mar-2014
Articles in Peer-reviewed Journals	Latchney SE, Rivera PD, Mao XW, Ferguson VL, Bateman TA, Stodieck LS, Nelson GA, Eisch AJ. "The effect of spaceflight on mouse olfactory bulb volume, neurogenesis, and cell death indicates the protective effect of novel environment." Journal of Applied Physiology. 2014 Jun 15;116(12):1593-604. http://dx.doi.org/10.1152/japplphysiol.01174.2013 ; PubMed PMID: 24744382; PubMed Central PMCID: PMC4152162 , Jun-2014

Articles in Peer-reviewed Journals	Walker AK, Rivera PD, Wang Q, Chuang JC, Tran S, Osborne-Lawrence S, Estill SJ, Starwalt R, Huntington P, Morlock L, Naidoo J, Williams NS, Ready JM, Eisch AJ, Pieper AA, Zigman JM. "The P7C3 class of neuroprotective compounds exerts antidepressant efficacy in mice by increasing hippocampal neurogenesis." Molecular Psychiatry. 2015 Apr;20(4):500-8. Epub 2014 Apr 22. http://dx.doi.org/10.1038/mp.2014.34 ; PubMed PMID: 24751964; PubMed Central PMCID: PMC4206684 (Reported originally in September 2014 as Advance online publication, April 22, 2014.) , Apr-2015
Articles in Peer-reviewed Journals	DeCarolis NA*, Rivera PD*, Ahn HF, Shih H-Y, Amaral WZ, LeBlanc JA, Malhotra S, Mukherjee S, Chen BPC, Eisch AJ. (*authors contributed equally and are listed alphabetically) "56Fe particle exposure results in a long-lasting increase in a cellular index of genomic instability and transiently suppresses adult hippocampal neurogenesis in vivo." Life Sciences in Space Research. 2014 Jul;2:70-9. http://dx.doi.org/10.1016/j.lssr.2014.06.004 ; PMID: 25170435 , Jul-2014
Articles in Peer-reviewed Journals	Latchney SE, Eisch AJ. "Therapeutic application of neural stem cells and adult neurogenesis for neurodegenerative disorders: regeneration and beyond." European Journal of Neurodegenerative Diseases. 2012;1(3):335-351. PubMed PMID: 25729743; PubMed Central PMCID: PMC4340249 , Jan-2012